

# **23CSE111 OBJECT ORIENTED PROGRAMMING**

## **S2 B.Tech CSE**

### **Department of Computer Science and Engineering**

### **Amrita School of Computing, Amritapuri Campus**

## **CASE STUDY**

### **PHASE – I (Analysis)**

#### **Problem Identification & Class Mapping (CO1)**

Group No :D2

Team Members :

| Roll No          | Student Name |
|------------------|--------------|
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#### **1. Problem Title :**

Online Food Delivery Management System

#### **2. Problem Description :**

Customers want convenient food ordering. Phone-based systems cause delays and errors. A centralized digital platform is needed. Modern lifestyles have increased demand for convenient food ordering. Customers prefer ordering through apps instead of visiting restaurants. However, phone-based ordering leads to miscommunication, delays, and incorrect deliveries. Restaurants struggle to manage multiple orders efficiently. Delivery coordination is another challenge. Without a structured system, delays and dissatisfaction occur. An Online Food Delivery System integrates customers, restaurants, and delivery agents on a single platform, ensuring smooth order processing and tracking.

#### **3. Identified Entities (Classes)**

| Class Name  | Description                                                                       |
|-------------|-----------------------------------------------------------------------------------|
| User        | Records personal details of customer like name, phone number.                     |
| Order       | It stores information about ordered items, total cost, order status, and receipt. |
| Restaurants | Stores restaurant information such as name, distance, rating.                     |

|          |                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|
| Delivery | It contains details like agent ID, contact information, and current location for tracking.               |
| Menu     | It contains details of items in restaurants such as item name, price, category, and availability status. |

#### 4. Class Attributes and Behaviours

| Class Name | Attributes                                                          | Methods(behaviours)                                                                                                                        |
|------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| User       | Username,<br>DeliveryAddress,<br>PhoneNumber,<br>Feedback,<br>Email | Login()<br>Feedback()<br>PaymentType()<br>setOrder()<br>setRestaurant()<br>checkUser()<br>displayUserDetails()                             |
| Order      | OrderID<br>OrderStatus<br>OrderedItems<br>Amount                    | CalculateAmount()<br>PlaceOrder()<br>CancelOrder()<br>ConfirmOrder()<br>saveOrder()<br>getOrderID()<br>printReceipt()<br>loadLastOrderID() |

|             |                                                                                     |                                                                                                           |
|-------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Restaurants | RestaurantName<br>Location<br>Rating<br>MenuItems<br>distance                       | ReceiveOrder()<br>AddMenuItem()<br>removeMenuItem()<br>BrowseRestaurants()                                |
| Menu        | ItemName<br>Price<br>Category<br>Availability                                       | DisplayItem()<br>displayMenu()<br>UpdatePrice()                                                           |
| Delivery    | AgentName<br>MobileNumber<br>CurrentLocation<br>VehicleNumber<br>AvailabilityStatus | AcceptDelivery()<br>UpdateLocation()<br>MarkDelivered()<br>TrackOrder()<br>MakePayment()<br>assignOrder() |